

Thermal Lab Report

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Group 10

Suzanne Volk

Lu Yu

Abhijith Sivaprasadan

1. Presentation of the lab setup

The main objective of this laboratory is to evaluate the performance and characterize a latent heat-based Mobile Thermal Energy Storage (M-TES) prototype.

The whole experimental system consists of oil bath, water cooling, coil heat exchanger (HX), heat transfer fluid (HTF) reservoir, pump, data logger, MTES and measurement system as shown in **Figure 1**.

Two thermostatic oil baths are used to regulate the temperature of the HTF, which is stored in a HTF reservoir. Heat transfer is facilitated by a coil HX, while a pump with a frequency controller circulates the HTF, maintaining the desired flow rate. Flow measurements are recorded using an omega flow meter. Temperature monitoring is performed using ten PT100 sensors placed at specific positions within the M-TES unit, complemented by a type-T thermocouple for tracking the HTF reservoir's temperature. The temperature data is logged via a Keithley data acquisition system connected to a computer, with pre-configured settings in Excel for data processing. Cooling is achieved using a cold water faucet when the buffer tank temperature falls below 100°C, allowing controlled heat discharge.

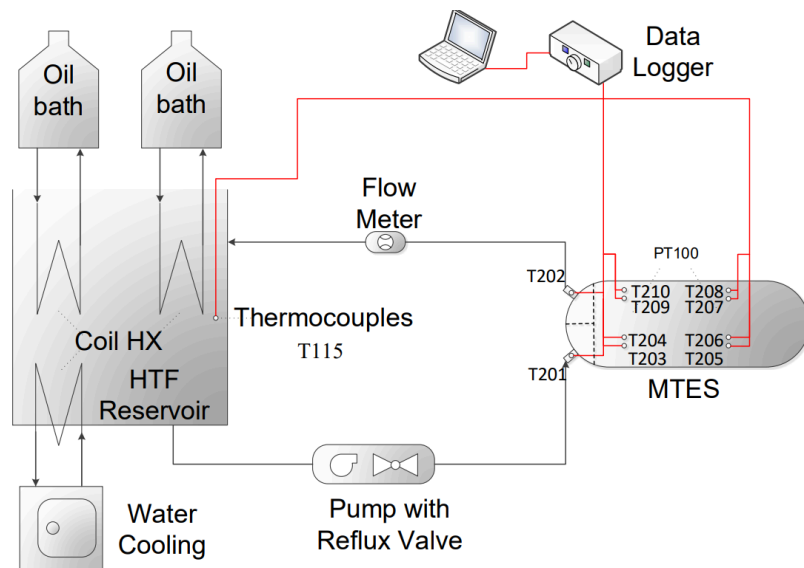


Figure 1 Experiment system patterns

2. Boundary conditions for the test

The boundary conditions for the test are established to simulate operational scenarios for the M-TES unit as shown in **Table 1**. Key parameters include the initial and final temperatures of the PCM and the HTF, along with controlled flow rates and ambient conditions to ensure consistent and reliable performance evaluation.

Additionally, the physical parameters of applied PCM and HTF are summarised and listed in **Table 2** and **Table 3**.

Table 1 Designed experimental set-up parameters

Parameter	Value
Empty shell weight (stainless steel)	150 kg
Ambient temperature	22.8 °C
HTF flow rate	2 L/min
Total volume of PCM	44 L
Starting temperature of PCM	71.1 °C
End temperature of PCM	40.1 °C

Table 2 PCM physical parameters

The PCM used is “**RT60 HC**”.

Parameter	Value
Nominal phase change temperature	60 °C
Density (liquid phase)	0.75 kg/L
Density (solid phase)	0.85 kg/L
Cp (specific heat capacity)	2 kJ/(kg*K)
Phase transition enthalpy	264.5 kJ/kg
Phase transition range solidification	62 °C → 51 °C
Total volume PCM	44 L

Table 3 HTF physical parameters

The HTF oil used is “**RENOLIN THERM 300 X**”.

Parameter	Value
Density (at 15 °C)	875 kg/m ³
Cp (at 50 °C)	2 kJ/(kg*K)

3. Theoretical calculations

Equation 1-4 presents the calculation methods for the theoretical storage capacity. Based on these equations and **Table 1-3**, the theoretical results were calculated and presented in **Table 4**.

$$Q_{PCM_{sensible\ solid}} = c_{P_{PCM\ solid}} * m_{PCM} * \Delta T_1 \quad \text{Eq.1}$$

$$Q_{PCM_{sensible\ liquid}} = c_{P_{PCM\ liquid}} * m_{PCM} * \Delta T_2 \quad \text{Eq.2}$$

$$Q_{PCM_{latent}} = L * m_{PCM} \quad \text{Eq.3}$$

$$Q_{storage} = Q_{PCM_{sensible\ solid}} + Q_{PCM_{sensible\ liquid}} + Q_{PCM_{latent}} \quad \text{Eq.4}$$

Table 4 Theoretical calculation results

Metric	Value
m_{PCM}	33 kg
$Q_{PCM_{sensible\ solid}}$	719.4 kJ
$Q_{PCM_{sensible\ liquid}}$	601.3 kJ
$Q_{PCM_{latent}}$	8728.5 kJ
$Q_{storage}$	10049.2 kJ

4. Experimental results

In this section we use the experimental data to calculate the heat that is transferred from the PCM to the HTF (oil). The temperature data is logged for ten sensors in the MTES, Chn 201 to Chn 210. 201 is at the inlet, 210 is at the outlet, as can be seen in figure 1. Chn 115 logs the temperature of the oil reservoir, to monitor the buffer tank HTF temperature. In figure 2, the experimental results are shown for all sensor temperatures over time. The total test lasted about 8 hours, the data was logged every 10 seconds.

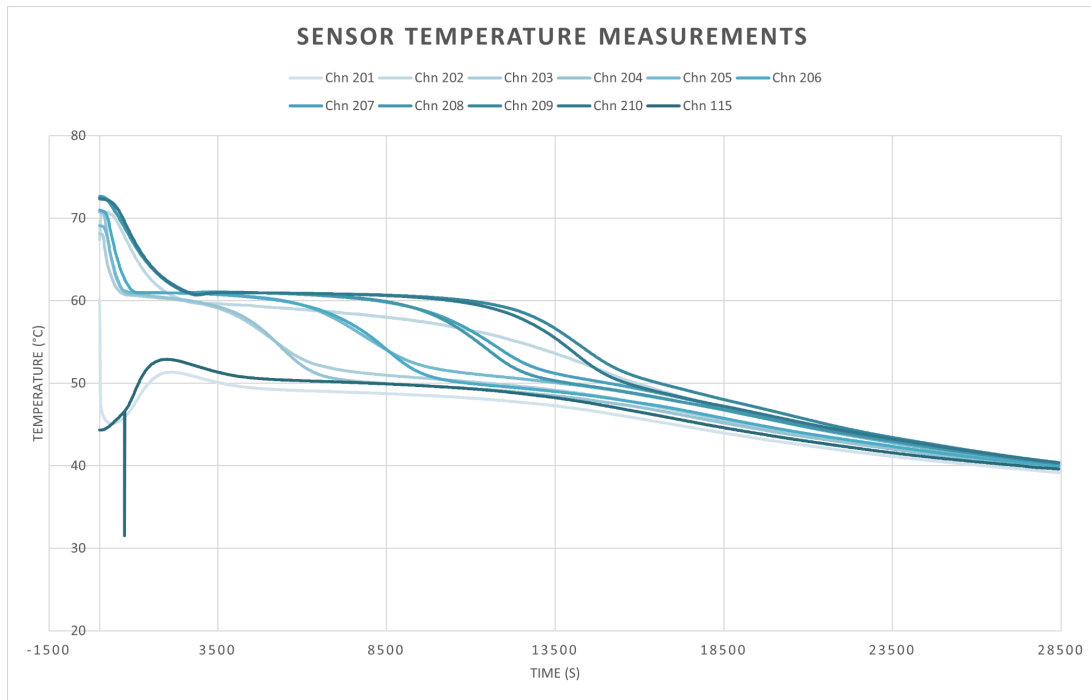


Figure 2 Sensor measurements vs Time

Analyzing this plot there are a few key observations:

- The inlet temperature at Chn 201 is the first to drastically decrease in temperature.
- The outlet temperature at Chn 202 follows a different type of course than the other sensors, it is steadily decreasing throughout the test.
- The pairs 203-204, 205-206, 207-208 and 209-210 follow a similar pattern, where the intermediate time between these consecutive drops is roughly 40 minutes.
- There is a sudden drop in the recorded temperature of the oil reservoir, which is most likely a measurement error.
- After 8 hours, all sensors log a temperature of 40 °C, as expected. This is the temperature of the oil baths, at this point no heat is extracted from the PCM anymore.

Method 1

To determine from the data the amount of heat transfer, we consider the difference between inlet and outlet temperatures (Chn 201 and Chn 202). The following formula is used to calculate Q_{HTF} , the experimental heat transfer.

$$Q_{HTF} = \sum_{t_1}^{t_{2850}} C_{p,HTF} \cdot \Delta T \cdot \dot{V} \cdot \rho \cdot \Delta t$$

where

$$- \Delta T = T_{Chn202} - T_{Chn201} \text{ (K)}$$

$$- \dot{V} = 2 \text{ L/min} = 2(\text{L/min}) \div 1000 (\text{L/m}^3) \div 60 (\text{s/min}) = 0.0000333 \text{ m}^3/\text{s}$$

- ρ and $C_{P,HTF}$ are functions of temperature, as can be seen in figures 3 and 4. To create these figures, the plots of the datasheet for the HTF were used. $C_{P,HTF}$ starts at a value of 2.033 kJ/kg K , at the end of the experiment this value has dropped to 1.945 kJ/kg K . Density ρ starts at 839.6 kg/m^3 and increases to 854.9 kg/m^3 at the end of the experiment.
- $\Delta t = 10 \text{ s}$

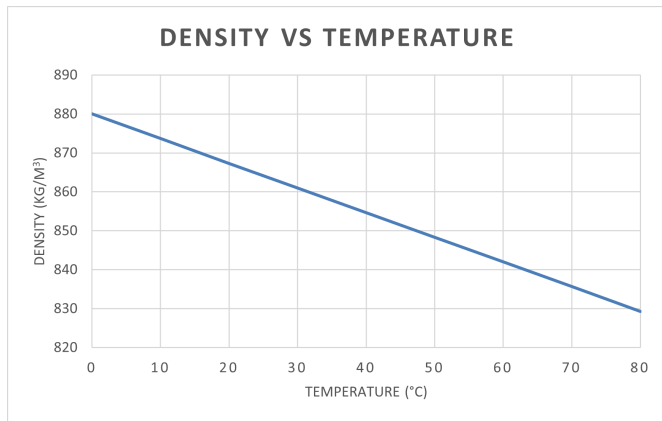


Figure 3: HTF density vs Temperature

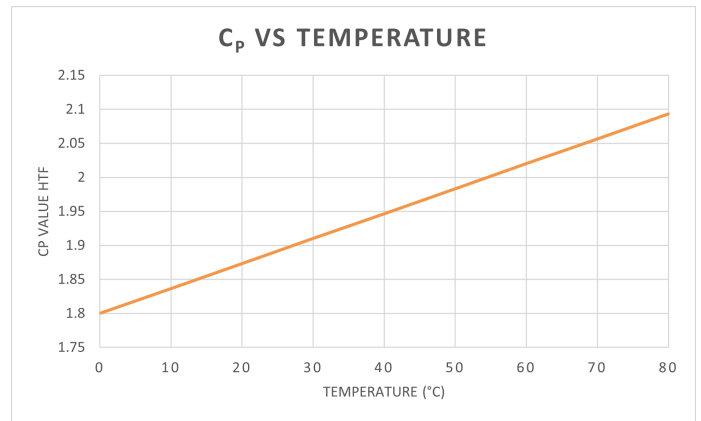


Figure 4: HTF Cp vs Temperature

Summing for all timesteps, while taking into account that ρ and $C_{P,HTF}$ vary, we obtain

$$Q_{HTF} = 10035.6 \text{ kJ}$$

Method 2

Another way to use this formula, is to take constant values for ρ and $C_{P,HTF}$. We take

$\rho = 825 \text{ kg/m}^3$ and $C_{P,HTF} = 2 \text{ kJ/kg K}$. Now summing again for all timesteps, we obtain

$$Q_{HTF} = 9814.6 \text{ kJ}$$

Method 3

Finally, we could also consider the fact that the heat from the PCM is not only transferred to the HTF, but also to the storage tank itself. For the tank we can assume $m = 150 \text{ kg}$, $C_p = 0.5 \text{ kJ/kg K}$, $\Delta T = 30 \text{ K}$.

The amount of heat taken up by the tank can thus be calculated as:

$$Q_{Tank} = m \cdot C_p \cdot \Delta T = 2250 \text{ kJ}.$$

Then, the total heat transfer can be calculated either as $10035.6 + 2250 = 12285.6 \text{ kJ}$ or as $9814.6 + 2250 = 12064.6 \text{ kJ}$

5. Analysis and discussion

The comparison between the theoretical calculations and experimental results provides valuable insights into the performance of the M-TES system:

Theoretical vs. Experimental Observations

1. **Theoretical Storage Capacity (Q_{storage}):**
 - The calculated value is **10,049.2 kJ**, considering the PCM's sensible and latent heat contributions.
2. **Experimental Heat Transfer (Q_{HTF}):**
 - **Method 1:** 10,035.6 kJ
 - **Method 2:** 9,814.6 kJ
3. **Including Tank Heat Absorption ($Q_{\text{HTF}} + Q_{\text{Tank}}$):**
 - **Method 1:** 12,285.6 kJ
 - **Method 2:** 12,064.6 kJ

Discrepancies and Contributing Factors

1. **Heat Absorption by the Stainless Steel Tank:**
 - The experimental setup includes a stainless steel tank weighing 150 kg. With an estimated specific heat capacity of 0.5 kJ/kg·K and a temperature difference of 30 K, the tank absorbs approximately 2,250 kJ of heat. This significant heat contribution was not accounted for in the theoretical model, leading to higher experimental values.
2. **Temperature-Dependent Material Properties:**
 - The theoretical calculations assume constant values for the specific heat capacity (C_p) and density (ρ) of the PCM and HTF. However, experimental data (Method 1) incorporates their temperature-dependent variations, resulting in more accurate heat transfer estimates.
3. **Heat Losses to the Environment:**
 - Despite efforts to control the experimental environment, minor heat losses to the surroundings could contribute to deviations between theoretical and experimental results.
4. **Measurement Uncertainties:**
 - Sensor inaccuracies, especially the sudden drop observed in the oil reservoir temperature, and the data logging frequency (10 seconds) may introduce errors in the experimental results.

Importance of Comprehensive System Modeling

This analysis underscores the importance of accounting for all system components, including the tank and HTF, and incorporating temperature-dependent properties.

Such comprehensive modeling is essential to accurately evaluate the performance of thermal energy storage systems.

6. Conclusion and future work

Conclusion

The latent heat-based Mobile Thermal Energy Storage (M-TES) system demonstrates a high degree of effectiveness in heat storage and transfer. Experimental results align closely with theoretical predictions when all system components, such as the PCM, HTF, and tank, are considered.

Key takeaways from this study include:

- The experimental heat transfer (10,035.6 kJ) aligns closely with the theoretical storage capacity (10,049.2 kJ) when focusing solely on the PCM.
- Including the tank's heat absorption increases the total heat transfer significantly (up to 12,285.6 kJ), surpassing theoretical predictions.
- Accurate modeling requires consideration of temperature-dependent material properties and contributions from auxiliary system components.

Future Work

To enhance the accuracy and applicability of M-TES systems, future studies should focus on the following areas:

- 1. Improving Theoretical Models:**
 - Incorporate dynamic, temperature-dependent properties of PCM and HTF into calculations.
 - Develop a comprehensive model that includes heat contributions from the tank and potential losses to the surroundings.
- 2. Enhancing Experimental Setup:**
 - Add more temperature sensors to capture detailed heat transfer dynamics across the tank and other system parts.
 - Perform experiments under varied flow rates, initial temperatures, and environmental conditions to generalize the findings.
- 3. Optimizing System Design:**
 - Investigate insulating materials or tank modifications to minimize heat absorption by the tank.
 - Explore improved heat exchanger designs to enhance heat transfer efficiency between the PCM and HTF.
- 4. Long-Term Performance Studies:**
 - Conduct repeated charge-discharge cycling tests to evaluate system durability and PCM stability over time.

- Analyze potential degradation effects on material properties and overall performance.

5. Scaling and Application Studies:

- Examine scalability for larger storage applications and its implications for system design.
- Validate models for predicting performance at different scales and operating conditions.

By addressing these areas, future research can contribute to the development of more efficient and scalable latent heat-based thermal energy storage solutions, fostering advancements in sustainable energy technologies.